BIT, SCREWDRIVER, PHILLIPS, RIBBED, CRUCIFORM, LONG SHANK, EXTERNAL HEXAGON SHANK, POWER DRIVEN

This specification sheet is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bits described herein shall consist of this specification and the latest issue of MIL-B-85643.

1. REQUIREMENTS:

1.1 Material: UNS T41901 (AISI S-1) or UNS T41902 (AISI S-2) tool steel in accordance with QQ-T-570, Type S1 or S2.

1.2 Dimensions and configuration:

Bit and retaining groove: In accordance with Figure 1 and Table I. Length in accordance with Table II.

Driver point: In accordance with Figure 1, Figure 2 and Table III.

1.2.1 Cruciform driver bit gage: In accordance with Figure 3 and Table IV.

1.3 Hardness: Rockwell 58 to 62 HRC.

1.4 Identification: Stamp military part number (see 1.8) and manufacturer's name, trademark or symbols.

1.5 Test recess: The test recess configuration and dimensions shall conform to Figure 4 and Table V.

1.6 Minimum ultimate torque loads: In accordance with Table I and MIL-B-85643.

1.7 Torque fatigue loads: In accordance with Table I and MIL-B-85643.

1.8 Example of military part number.

M85643/2 - 2 - 188 - A

- A — Indicates larger "OD" dimension when desired (see Tables I and II)
- 188 — Indicates length "N" (see Table II)
- 2 — Driver size number (see Table I)
- M85643/2 — Basic specification sheet number

AMSC N/A FSC 5120

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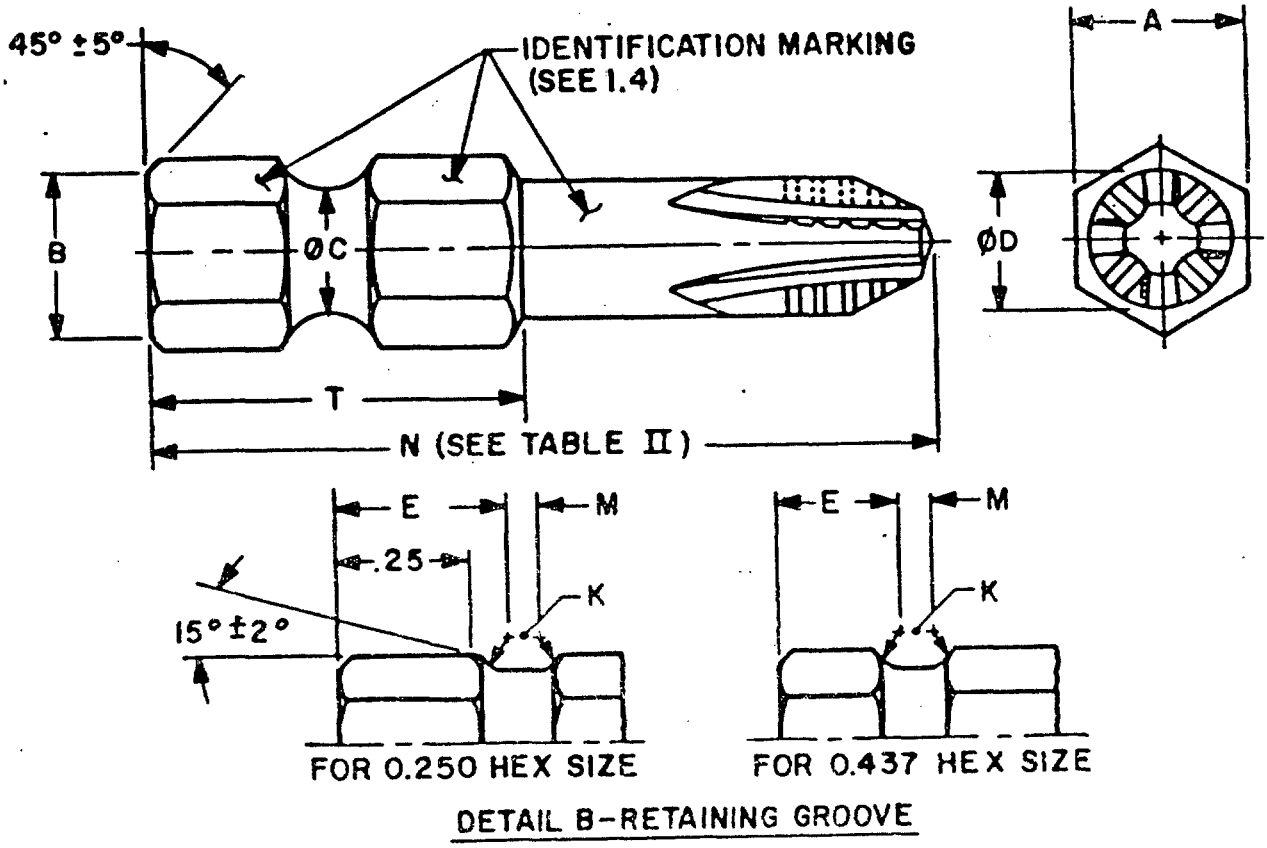


FIGURE 1. Bit.

TABLE I. Bit sizes and dimensions.

DRIVER SIZE NUMBER	A HEX SIZE +.000 -.004	ØB REF	ØC ±.010	ØD SHANK DIA	E ±.020	K RAD	M ±.020	T	MIN ULT TORQUE (LBF-IN.)	FATIGUE TORQUE (LBF-IN.
02	.250	.203	.188	.185 .180	.375	.094	.062	1.375 .625	90	45
02A				.247 .242						
03	.437	.391	.344	.434 .429	.250	.109	.047	1.375 1.000	325	160

Dimensions are in inches.

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TABLE II. Overall length and dash number.

Driver Size Number	Recess Size	Screw Dia Size Range for Recess Size Ref.	A Hex Size Ref.	OD Shank Dia Ref.	N+.12				Shank Dia
					1.88	2.75	3.50	6.00	
					Dash Number Indicating Driver Size, Bit Length & Shank Dia				
02	2	#6, #8, #10	.250	.185	2188	2275	2350	2600	Std
02A				.247	2188A	2275A	2350A	2600A	Large
03	3	1/4	.437	.434	3188	3275	3350	3600	Std

All dimensions in inches.

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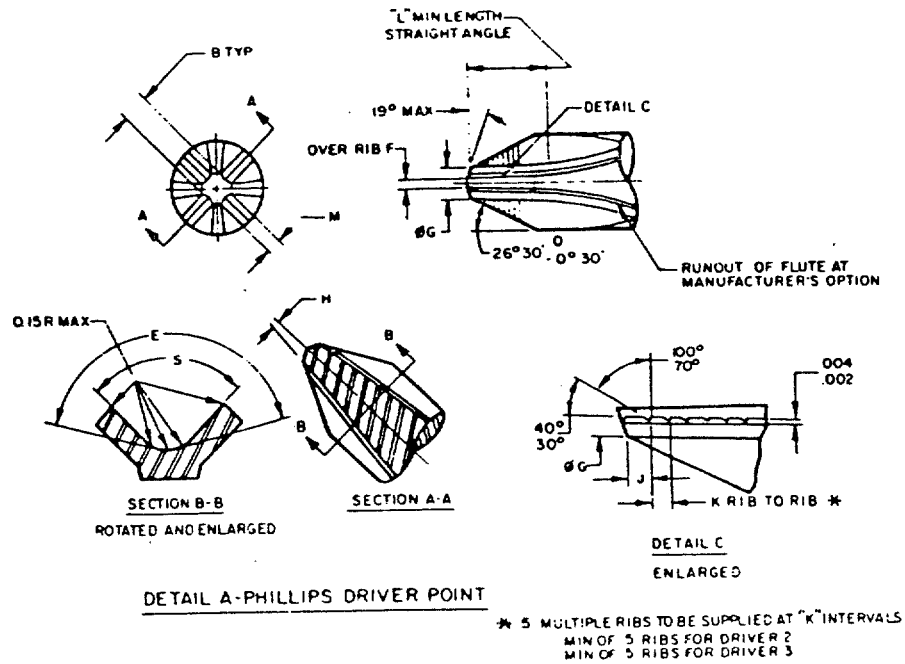


FIGURE 2. Driver point.

TABLE III. Phillips Driver Point Dimensions for ribbed cruciform configuration.

Driver Size	2	3
B ±.0010	.0606	.0983
E +0°30' 0	140°0'	146°0'
F Over Rib	.023-.025	.029-.031
G ±.001	.090	.150
H 0 -0°30'	5°45'	5°45'
J Max	.032	.045
K	.015-.022	.015-.022
L Min	.188	.281
M ±.001	.0434	.0826
S +0°30' 0	92°0'	92°0'

All dimensions in inches.

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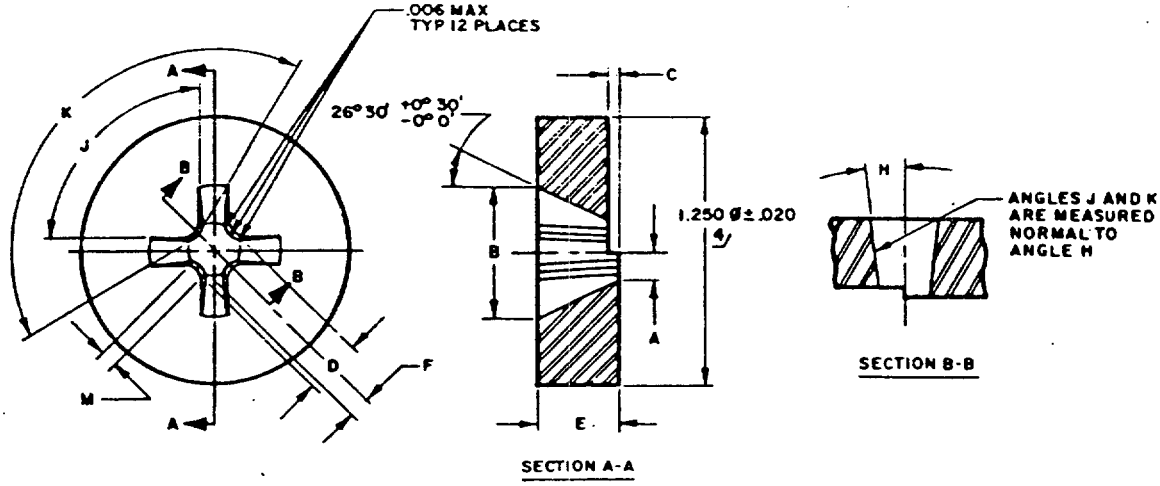


FIGURE 3. Driver point gage.

TABLE IV. Dimensions for cruciform Phillips driver point gage.

Dash Number Driver Point Size	2	3
A ±.0002	.0455	.0755
EF	.278	.432
C ±.001	.010	.010
D Ref	.098	.156
E +.000 -.015	.188	.281
F +.0004 -.0000	.0303	.0496
H +0°15' -0°0'	5°45'	5°45'
J +0°0' -0°15'	92°0'	92°0'
K +0°0' -0°15'	140°0'	146°0'
M ±.001	.032	.079

All dimensions in inches.

NOTES:
1. Material: CRES shall conform to Class 440C per QQ-S-763 (UNS S44004).
2. Hardness: Rockwell 58-62 HRC.
3. Marking: Identify top surface with M85643, phrase "Driver Point Size" and applicable dash number indicating driver point size.
   Ex: M85643
   Driver Point Size 2
4. The "G" dia of the driver point is to fall between the limits of "C" on this gage.

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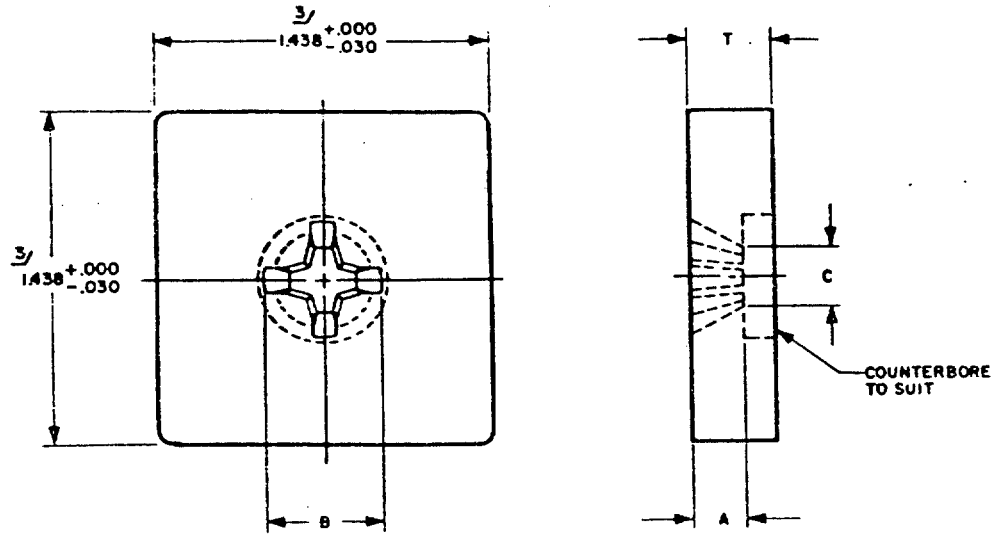


FIGURE 4. Test recess.

TABLE V. Dimensions for test recess for testing cruciform Phillips bits.

Recess Size	A Ref.	B ±.002	C ±.002	T ±.002
2	.143	.233	.090	.375
3	.236	.386	.150	

All dimensions in inches.

NOTES:
1. Material: Tool steel shall conform to Type S2 per QQ-T-570 (UNS T41902).
2. Hardness: Rockwell 62 to 64 HRC.
3. Size can be adjusted to suit if needed.
4. Marking: Identify top surface with M85643 and recess size.
   Example: M85643
   Recess Size 2

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1.9 ACR and Phillips are registered trademarks of Phillips Screw Company. Patent 4,187,892 expires 12 February 1997. The Government does not have a royalty-free license.

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